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RESEARCH ARTICLE

A Self-Supervised Monocular Depth Estimation Framework Based on Detail Recovery and Feature Fusion

SHUN LI[1], CHONGZHENG HUANG[2], XIANGZHE LI[1], AND ZHENGYOU LIANG[1,3]
[1]School of Computer, Electronics and Information, Guangxi University, Nanning 530004, China
[2]School of Information Engineering, Guangxi Polytechnic of Construction, Nanning 530007, China
[3]Guangxi Key Laboratory of Multimedia Communications and Network Technology, Nanning 530004, China

Corresponding author: Zhengyou Liang (zhyliang@gxu.edu.cn)

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ABSTRACT With the rapid development of autonomous driving and robotics, the demand for high-precision depth estimation is growing steadily. The emergence of self-supervised learning methods has reduced the reliance on expensive and hard-to-obtain ground truth data. However, existing models still face challenges in design, such as insufficient detail recovery, inadequate feature fusion, and limited global contextual understanding, which lead to deficiencies in the accuracy and precision of the generated depth maps. To address these challenges, this paper proposes a novel self-supervised monocular depth estimation network, Enhanced Depth Fusion Net (EDFNet). Based on the encoder-decoder architecture, EDFNet introduces three key modules: the Adaptive Selective Attention Module (ASAM), the Multi-Dimensional Attention Fusion Module (MDAFM), and the Hybrid Parallel Convolution Block (HPCB). Specifically, ASAM selectively emphasizes critical features and spatial locations in images to enhance the model's ability to capture details. MDAFM captures feature information from different perspectives and levels to address the problem of critical information loss, while HPCB effectively captures and fuses local and global information through parallel processing. The synergy of these modules enables EDFNet to generate high-quality depth maps in various complex scenarios. Experimental results on the KITTI dataset demonstrate that the proposed method significantly outperforms state-of-the-art approaches across multiple evaluation metrics, including Abs Rel, Sq Rel, RMSE, $\delta < 1.25$, effectively mitigating the issues of detail loss and edge blurring in depth maps. Additionally, EDFNet demonstrates excellent generalization performance on the Make3D, NYU Depth V2, and Cityscapes datasets.

INDEX TERMS Detail recovery, feature fusion, monocular depth estimation, self-supervised learning.

I. INTRODUCTION

Monocular depth estimation, which aims to infer the 3D structure of a scene from a single image, is one of the most important research directions in the field of computer vision. This task is of great significance in practical applications such as autonomous driving [1] robot navigation [2], [3], [4], augmented reality (AR), and virtual reality (VR). However, monocular depth estimation also faces many challenges, including the difficulty of detecting small objects [5], loss of details and edge blurring [5], and problems with dynamic scenes and occlusions [6]. These issues often make it difficult for existing methods to recover the details of small objects when dealing with complex scenes, resulting in blurry edges in the generated depth maps.

In the field of monocular depth estimation, although supervised learning methods have achieved significant results, their dependence on a large amount of labeled data [7], [8] not only increases the cost of data collection and processing but also limits the promotion of models in practical applications. Consequently, self-supervised learning methods have gradually become an effective way to solve the

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problem of monocular depth estimation. Self-supervised learning trains depth estimation models without relying on real depth labels by utilizing the inherent structure of images or introducing external constraints. DispNet [9] and Monodepth2 [7] have significantly improved the accuracy of monocular depth estimation by introducing disparity consistency loss or multi-scale feature fusion techniques.

The design of monocular depth estimation networks mainly uses cutting-edge technologies such as encoder-decoder [9] architecture, attention mechanisms, multi-feature fusion, and self-supervised learning frameworks. The encoder-decoder architecture is one of the core designs of monocular depth estimation networks. the encoder is tasked with extracting spatial features from the input image, while the decoder is responsible for transforming these features into a depth map. HR-Net [10] uses the encoder-decoder architecture as its foundation, and its encoder can capture spatial features at different scales, thereby improving the model's adaptability to complex scenes. Attention mechanisms play an important role in improving the accuracy of depth estimation. By using spatial attention and channel attention modules, the model can focus on key regions or feature channels in the image, thereby enhancing the model's ability to capture details. DIFFNet [11] combines multi-resolution feature fusion and spatial attention modules, significantly reducing artifacts and improving accuracy. Multi-scale feature fusion technology can effectively alleviate the problem of detail loss in depth estimation. By capturing image detail information at multiple scales, the model can retain more details when generating depth maps. MSFNet [12] incorporates an Upsample-Stage Fusion (USF) module into its network architecture to achieve high-quality depth map generation. Self-supervised learning frameworks train models by introducing additional constraints such as disparity consistency and occlusion masks. These constraints not only enhance the model's adaptability to dynamic scenes but also improve the accuracy and robustness of depth estimation. DevNet [6] effectively solves the impact of occlusion problems on depth estimation in dynamic scenes by using occlusion regularization technology and predicting the occlusion probability density of sampling points.

In recent years, although significant progress has been made in the field of monocular depth estimation [13], [14], [15], problems such as insufficient recovery of depth map detail information and the presence of artifacts still exist. The proposed Enhanced Depth Fusion Net (EDFNet) draws upon the essence of previous research [7], [16] and innovatively expands upon it. Specifically, we introduce three key modules: the Adaptive Selective Attention Module (ASAM), the Multi-Dimensional Attention Fusion Module (MDAFM), and the Hybrid Parallel Convolution Block (HPCB), enabling our model to generate more complete contours and clearer details. Its optimization design in encoder-decoder architecture, attention mechanism, and multi-scale feature fusion enables it to generate higher quality depth maps when dealing with complex scenes, improving the accuracy of depth estimation and providing new ideas for solving the above problems.

The main contributions of this work can be summarized as follows:

1. We propose a new monocular depth estimation network, EDFNet, which extends the existing feature fusion theory by introducing a multi-dimensional attention mechanism and an adaptive selection mechanism, providing new ideas for solving the problems of detail loss and edge blurring in monocular depth estimation depth maps.

2. By combining the Adaptive Selective Attention Module (ASAM), the Multi-Dimensional Attention Fusion Module (MDAFM), the Hybrid Parallel Convolution Block (HPCB), and Skip-connection [17], EDFNet achieves significant optimization in the encoder-decoder architecture, enabling it to better tackle depth information recovery in complex scenes, thereby providing a new direction for the architectural design of deep learning models in monocular depth estimation.

3. On a single NVIDIA RTX 4060ti (16GB), we demonstrated the network's superior accuracy through the KITTI [18] dataset and further explored the model's generalization ability on the Make3D [19] dataset, the NYU Depth V2 [20] dataset, and Cityscapes dataset [21]. In addition, we conducted ablation experiments to systematically evaluate the effectiveness of different design choices.

II. RELATED WORD

Estimating the depth from a single image is an ill-posed problem due to the lack of reliable cues. Traditional geometric methods for obtaining depth information from a single image are often challenging. To address this challenge, researchers have increasingly explored methods using deep learning, which can be broadly categorized into two types: supervised monocular depth estimation and self-supervised monocular depth estimation.

A. SUPERVISED MONOCULAR DEPTH ESTIMATION METHODS

Supervised monocular depth estimation methods rely on datasets with real depth information, training deep neural networks to learn the mapping relationship from the input image to the estimated depth. Eigen et al. [22] were among the first to employ deep networks for estimating depth maps from single images. They improved certainty and accuracy by using a stack of two depth networks to perform coarse global predictions followed by refinement. Fu et al. [23] adopted a multi-scale network architecture and transformed the depth estimation task into an ordered regression problem, effectively enhancing the model's accuracy and convergence rate. Subsequently, supervised depth learning methods have been continuously improved [24], [25]. However, due to issues such as poor generalization and difficulty in obtaining real depth values, supervised methods have encountered bottlenecks, prompting researchers to explore self-supervised monocular depth estimation.

B. SELF-SUPERVISED MONOCULAR DEPTH ESTIMATION METHODS

Self-supervised monocular depth estimation methods do not require the use of real depth data labels. Instead of relying on expensive depth sensors to collect ground truth depth data, they use other information such as image geometric consistency or temporal relationships as supervisory signals to achieve the ability to predict depth maps from a single image. Godard et al. [7] proposed the Monodepth2 model, which uses minimum reprojection error and automatic masking error to solve problems with occlusion and dynamic objects, achieving state-of-the-art results at the time. For dynamic class (DC) objects, such as moving people or cars, Marvin Klingner et al. [26] designed the SGDepth model, which, based on self-supervision, uses cross-domain training, semantic masking, and dynamic object frame selection methods to demonstrate efficiency in benchmark tests. In the same year, Johnston et al. [27] expanded the attention mechanism on the basis of Monodepth [28], enabling the exploration of more general contextual information. By using discrete disparity prediction, they were able to estimate depth uncertainty and provide more robust and clearer depth estimation than common continuous disparity prediction. The CADepth-Net [29] also achieved state-of-the-art performance in depth estimation tasks, introducing a structure-aware module and a detail-emphasis module to achieve better scene understanding and richer feature representation, thereby improving the precision and clarity of depth prediction and solving performance bottlenecks and artifact problems caused by insufficient scene structure modeling and improper detail handling.

C. NETWORK ARCHITECTURE FOR MONOCULAR DEPTH ESTIMATION

The design of effective network architectures plays a crucial role in achieving good results in monocular depth estimation. Following the introduction of AlexNet by Krizhevsky et al. [30], which significantly improved image recognition accuracy, Convolutional Neural Networks (CNNs) have remained a dominant model in computer vision, particularly with the VGG [31] model being widely used in object detection, semantic segmentation, and depth estimation applications. The ResNet model [17], introduced a few years later, addressed the vanishing gradient problem in training deep networks through residual learning, making it possible to train deeper networks and achieve better results. The GeoNet framework [32] learns jointly in an end-to-end manner, extracting geometric relationships from the predictions of various modules and combining them into an image reconstruction loss for reasoning about static and dynamic scene components. The DVO model [33] proposed that depth CNN predictors could learn without pose CNN predictors, significantly achieving improved performance through a novel depth normalization strategy. Concurrently, optimizations in CNN architectures have also positively impacted training parameters. The HR-DepthNet [6] architecture improves high-resolution estimation performance by restructuring depth network skip connections and applying feature fusion techniques. The RAFM architecture [34] integrates dilated convolution modules of varying rates into the current recurrent multi-scale refinement system (R-MSFM), constructing a powerful monocular depth estimation system.

Despite the outstanding performance of CNN in computer vision tasks, they struggle to capture long-range dependencies due to the limitations of convolutional operations. TransDepth [35] was the first to utilize a Vision Transformer (ViT) as the encoder for monocular depth estimation networks. The MonoFormer architecture proposed by Bae et al. [36] employs a hybrid CNN-Transformer network with multi-level feature aggregation. Kim and Shim [37] adopted the Swin Transformer as a feature extractor to capture local geometric features and global semantic features, designing a Dense Cascade Multi-scale Network (DCMNet) that directly connects feature maps of different scales through a top-down cascading path, enhancing interconnections between decoding layers and generating high-quality multi-scale depth outputs. The Lite-Mono network architecture proposed by Zhang et al. [16] is a more advanced network under the CNN-Transformer framework, incorporating a Continuous Dilated Convolution (CDC) module and a Local-Global Feature Interaction (LGFI) module to extract rich multi-scale local features while effectively balancing model complexity and inference speed. In terms of improvements to the encoder-decoder structure, RSANet [8] employs an interleaved depth skip connection structure with a new decoder, vertical concatenation modules, and scale recovery modules to enhance the accuracy of high-resolution image depth predictions and model generalization capabilities. Recently, Sun et al. [38] integrated a hybrid transformer with a lightweight pose network, context-prior attention mechanisms, and semantic boundary loss to enhance the representational capability and generalization performance of depth estimation by incorporating various prior information. Although these methods have made significant progress in certain areas, they still face numerous challenges and limitations. For instance, existing models often struggle to accurately reconstruct fine structures and object edges in images, leading to depth maps that lack detail richness. In terms of feature fusion, models may fail to fully capture the relationship between global contextual information and local details when integrating multi-scale features, thereby affecting the accuracy and consistency of monocular depth estimation. In addition, existing models often struggle to maintain stable performance across diverse scenarios. This instability limits their reliability and broad applicability, especially in safety-critical areas like autonomous driving and robot navigation. Consequently, further exploration of effective information extraction from scenes to enhance the performance of monocular depth estimation methods remains a key direction for future research.

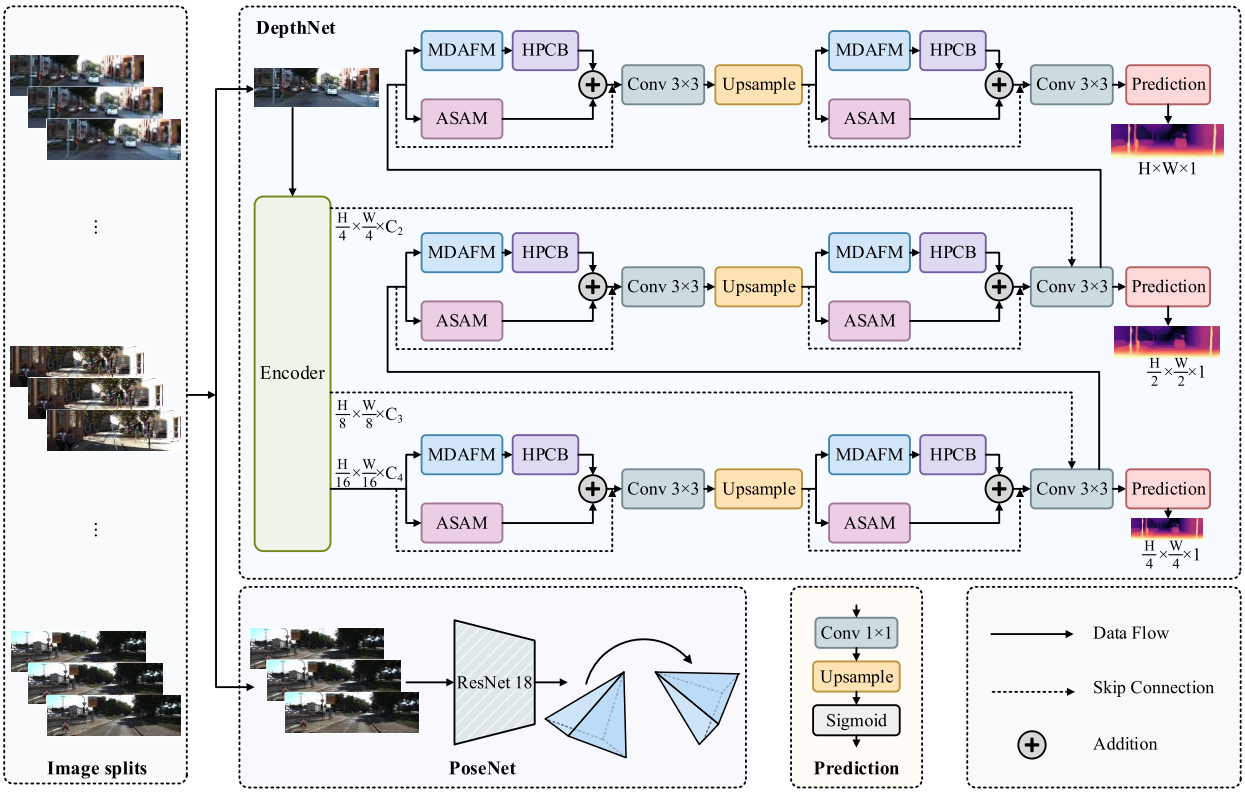


FIGURE 1. Overview of the proposed EDFNet architecture. It adopts a dual-mode design, consisting of DepthNet for extracting depth information and PoseNet [7], [16], [28] for estimating camera pose information. To effectively capture and fuse detail information, the DepthNet-decoder utilizes the Adaptive Selective Attention Module (ASAM), the Multi-Dimensional Attention Fusion Module (MDAFM), and the Hybrid ParallelConvolution Block (HPCB) to process the multi-scale features extracted by the DepthNet-encoder.

III. METHODS

A. MOTIVATION

Currently, self-supervised monocular depth estimation network architectures are rapidly advancing. The flexibility of the encoder-decoder architecture allows it to adapt to various task requirements [7], [8]. An excellent decoder can enhance the model's focus on details when processing features from the encoder, facilitating more effective detail recovery and feature fusion [23], thereby ensuring that the generated depth maps are more stable and accurate. Based on this analysis, we designed the self-supervised monocular depth estimation model EDFNet, aimed at achieving high-precision depth prediction. **Figure 1** illustrates the basic structure of the proposed monocular depth estimation model, which consists of a DepthNet and a PoseNet [7], [16], [28]. This model performs depth estimation for each input image and combines the existing PoseNet and loss functions for self-supervised training. The following sections provide a detailed description of the architecture.

B. DEPTHNET

1) DEPTHNET-ENCODER

Inspired by [16], we introduce the Lite-Mono-encoder as the DepthNet-encoder. The DepthNet-encoder adopts a four-stage process for multi-scale feature aggregation. In the first stage, the input image of size $H \times W \times 3$ is fed into the convolution stem for extracting features and downsampling, generating feature maps of size $\frac{H}{2} \times \frac{W}{2} \times C_1$. The second stage consists of a downsampling layer, Consecutive Dilated Convolutions (CDC) [16] module, and the Local-Global Features Interaction (LGFI) [16] module. The feature maps from the previous stage and the feature maps obtained by averaging the original image are used as the input for this stage. This structure aims to reduce the loss of spatial information due to downsampling, and the output feature maps size are $\frac{H}{4} \times \frac{W}{4} \times C_2$. The third and fourth stages adopt the same structure as the second stage, resulting in feature maps of sizes $\frac{H}{8} \times \frac{W}{8} \times C_3$and $\frac{H}{16} \times \frac{W}{16} \times C_4$ respectively.

2) DEPTHNET-DECODER

In the decoder design, we skillfully integrated a variety of advanced technologies to maximize the retention of features at each scale and enhance the understanding of detail information and global context during feature fusion, thereby generating high-quality depth maps. MDAFM and ASAM serve as the initial feature extraction core of the decoder. The former is responsible for integrating features at different scales from multiple perspectives, using local, global, and serial convolution branches to capture multi-dimensional spatial information from the input features. The latter

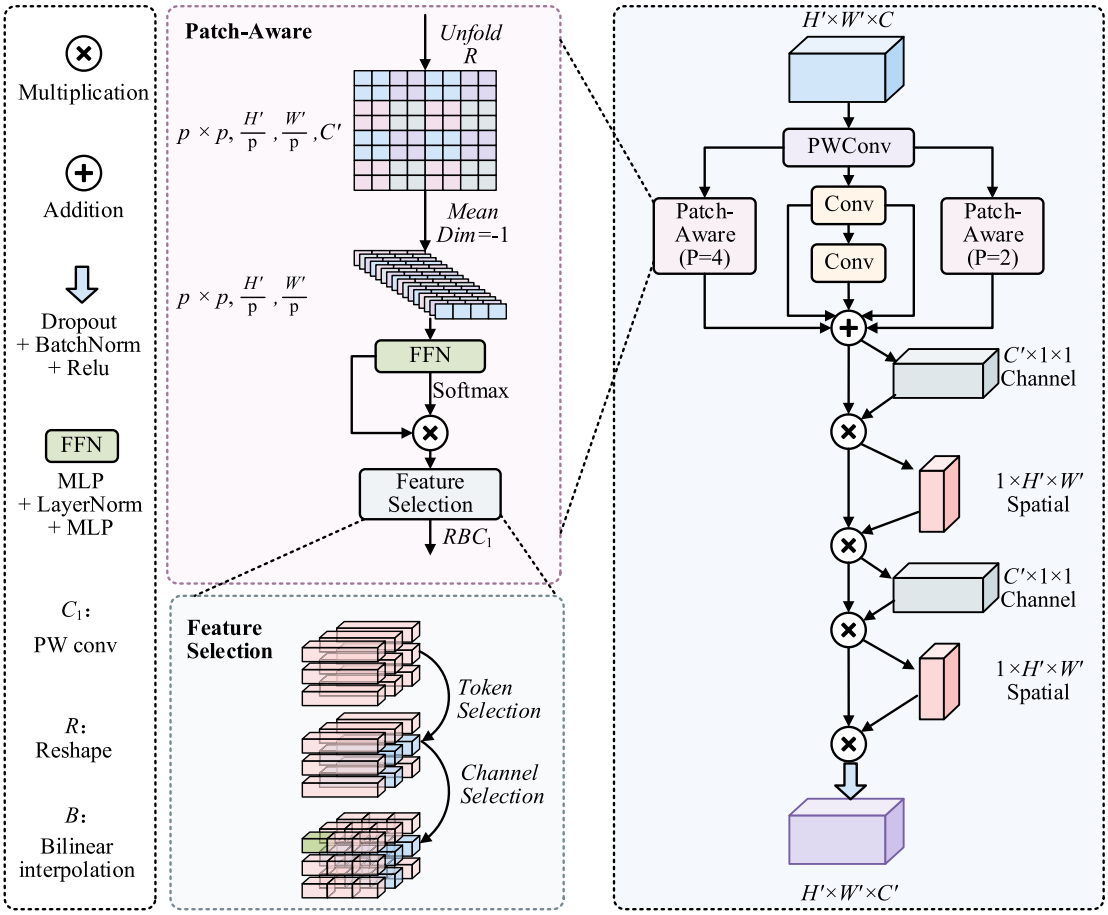


FIGURE 2. Structure of Multi-Dimensional Attention Fusion Module (MDAFM).

selectively emphasizes key features and spatial locations through adaptive selection of convolution kernel sizes and attention mechanisms, based on the characteristics of the input features. HPCB, which takes the output of MDAFM, is a key component in the decoder responsible for deep feature fusion and optimization. It is designed with four parallel branches, each digging into and integrating features from different angles. In addition, the decoder also employs element-wise addition fusion, Skip-connection [17], convolutional refinement, and upsampling techniques. Element-wise addition fusion adds the features processed by HPCB and ASAM element-wise, deeply fusing the feature information from different modules, retaining their advantages, and enhancing feature expression capabilities. Skip-connection directly passes low-level feature information to higher levels, avoiding the vanishing gradient problem caused by increased network depth. Convolutional refinement further processes the fused features through a 3×3 convolution, extracting richer and more delicate semantic information. Upsampling technology is used to improve the spatial resolution of the feature maps, gradually transitioning from low-resolution depth maps to high-resolution depth maps, and finally generating depth maps aligned with the original image resolution. We set up Prediction Head to output inverse depth maps at full resolution, 1/2 resolution, and 1/4 resolution respectively.

C. MULTI-DIMENSIONAL ATTENTION FUSION MODULE (MDAFM)

The MDAFM replaces traditional convolution operations in the basic components of the decoder, effectively addressing the issue of critical information loss. As illustrated in **Figure 2**, MDAFM employs a parallel multi-branch structure, comprising local, global, and serial convolution branches. Each branch captures information from multiple perspectives and levels. Additionally, we utilize attention mechanisms to perform adaptive feature enhancement, with the attention part employing two channel attention [39] and two spatial attention [40] in series. In the multi-branch feature extraction process, the input feature tensor $F \in R^{H' \times W' \times C}$ is first processed through point-wise convolution to obtain $F' \in R^{H' \times W' \times C'}$. Subsequently, the local, global, and serial convolution branches compute the following features: $F_{local} \in R^{H' \times W' \times C'}$, $F_{global} \in R^{H' \times W' \times C'}$, and $F_{conv} \in R^{H' \times W' \times C'}$, respectively. The three branches' outputs are then summed to produce $\tilde{F} \in R^{H' \times W' \times C'}$. The patch size

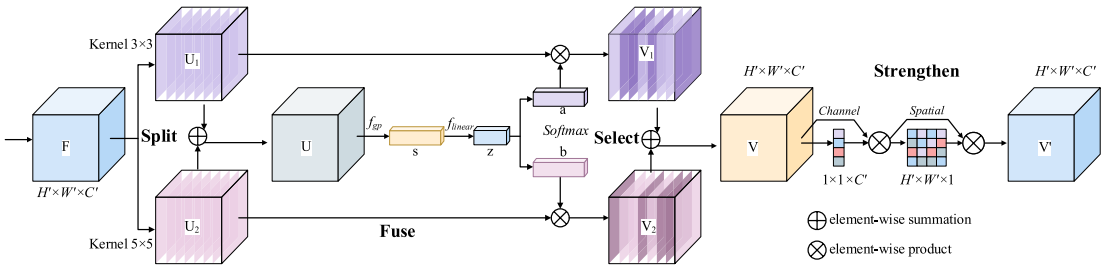


FIGURE 3. Structure of Adaptive Selective Attention Module (ASAM).

parameter p is determined by aggregating and displacing non-overlapping patches in the spatial dimension, with local and global branches differing in their patch size parameters—set to 2 and 4, respectively. Calculating the attention matrix between non-overlapping patches is crucial for facilitating local and global feature extraction and interaction. Initially, we employ Unfold and reshape operations, segmenting F' into a series of contiguous blocks of size $\left(p \times p, \frac{H'}{P}, \frac{W'}{P}, C'\right)$. Each channel is then averaged to obtain $\left(p \times p, \frac{H'}{P}, \frac{W'}{P}\right)$, followed by linear computation using (FFN) [41]. An activation function is applied to derive the probability distribution of the linear computed features in the spatial dimension.

The feature selection [42] is an effective strategy for obtaining weighted results. Initially, the weighted results are represented as $(t_i)_{i=1}^{C'}$, where $t_i \in R^d$ denotes the i-th output label, and d is defined as $\left(\frac{H' \times W'}{P \times P}\right)$. The feature selection process for each label yields the output $\hat{t}_i = P \cdot sim(t_i, \varepsilon) \cdot t_i$, where $P \in R^{C' \times C'}$ is task-specific parameters, $sim(\cdot, \cdot)$ is the cosine similarity function, and $\varepsilon \in R^{C'}$ serves as task embedding. Each label t_i is re-weighted based on its correlation with the task embedding. We then apply a linear transformation of P for channel selection, along with reshaping and interpolation operations, to obtain the local branch features $F_{local} \in R^{H' \times W' \times C'}$ and global branch features $F_{global} \in R^{H' \times W' \times C'}$. Finally, two 3×3 convolution layers are used to construct the serial convolution, resulting in $F_{conv1} \in R^{H' \times W' \times C'}$ and $F_{conv2} \in R^{H' \times W' \times C'}$, which are summed to produce the serial convolution branch features $F_{conv} \in R^{H' \times W' \times C'}$.

After extracting features through the local, global, and serial convolution branches, we employ efficient channel attention and spatial attention for adaptive feature enhancement, providing more comprehensive and effective feature extraction capabilities. Consequently, the final output of MDAFM, denoted as $F'' \in R^{H' \times W' \times C'}$, is derived from $\tilde{F} \in R^{H' \times W' \times C'}$ processed through two one-dimensional channel attention maps $M_c \in R^{1' \times 1' \times C'}$ and two two-dimensional spatial attention maps $M_s \in R^{H' \times W' \times 1}$, as follows:

$$F_c = M_c\left(\tilde{F}\right) \otimes \tilde{F}, F_s = M_s(F_c) \otimes F_c \tag{1}$$

$$F_c' = M_c(F_s) \otimes F_s, F_s' = M_s\left(F_c'\right) \otimes F_c' \tag{2}$$

$$F'' = \delta\left(\mathrm{B}\left(dropout\left(F_s'\right)\right)\right) \tag{3}$$

where $\otimes$ denotes element-wise multiplication, F_c and F_c' are features selected through channel attention, F_s and F_s' are features selected through spatial attention, $\delta(\cdot)$ denotes the ReLU function [43], and $B(\cdot)$ represents Batch Normalization [44].

D. ADAPTIVE SELECTIVE ATTENTION MODULE (ASAM)

The ASAM selectively emphasizes critical features and spatial locations by adaptively adjusting convolution kernel selection [45] and incorporating attention mechanisms. This approach provides richer feature representations and achieves greater flexibility and accuracy in preserving and enhancing mage details. As illustrated in **Figure 3**, ASAM is implemented through four operations: Split, Fuse, Select, and Strengthen. In the Split operation, given an input feature map $F \in R^{H' \times W' \times C'}$, we apply two convolution kernels of size 3×3 and 5×5, respectively, to produce intermediate feature maps $U_1 \in R^{H' \times W' \times C'}$ and $U_2 \in R^{H' \times W' \times C'}$. These two branches enable the model to capture multi-scale information. Next, in the Fuse operation, the outputs from the two branches are combined via element-wise summation, effectively integrating information from different receptive fields:

$$U = U_1 + U_2 \tag{4}$$

where $U \in R^{H' \times W' \times C'}$ retains the same dimension as the input feature maps. This is followed by global average pooling along the spatial dimensions $H \times W$ yielding a one-dimensional vector s_c that represents the importance of each channel:

$$s_c = f_{gp}(U_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} U_c(i, j) \tag{5}$$

To improve computational efficiency, we define a compressed channel dimension d:

$$d = max\left(\frac{C}{r}, L\right) \tag{6}$$

where C denotes the number of channels in the input feature map, r represents the hyperparameter reduction, and L indicates the minimum value of d, which is set to 16. Next, a linear transformation is used to project the original C dimensional information into $z \in R^{d \times 1}$ for information extraction. Subsequently, two linear transformations are applied to transform $z \in R^{d \times 1}$ back to the original

C dimensions, thus completing the information extraction for the channel dimension:

$$z = f_{linear}(s) = \delta(B(W_s)) \tag{7}$$

where $\delta(\cdot)$ denotes the ReLU function [43], $B(\cdot)$ denotes Batch Normalization [44], and $W \in R^{d \times C}$. Then the Select operation is performed, using the Softmax function for normalization. At this point, each channel corresponds to a score, representing the importance of that channel, effectively serving as a scoring mask:

$$a_c = \frac{e^{A_c z}}{e^{A_c z} + e^{B_c z}}, b_c = \frac{e^{B_c z}}{e^{A_c z} + e^{B_c z}} \tag{8}$$

where a and b represent the soft attention of U_1 and U_2, respectively. $A, B \in R^{C \times d}$, $A_c \in R^{1 \times d}$ is the c-th row of A and a_c is the c-th element of a, likewise B_c and b_c. The obtained masks are multiplied by the corresponding U_1 and U_2, to get V_1 and V_2, respectively. Then, the two features are added together to perform information fusion, resulting in the feature map V. Compared to the initial F, the feature map V has undergone information purification and contains multi-scale information:

$$V = a_c \cdot U_1 + b_c \cdot U_2, a_c + b_c = 1 \tag{9}$$

Finally, in the Strengthen operation, the feature map V combines channel and spatial attention mechanisms, enhancing the feature's expressiveness by weighting the features of different channels to automatically identify and enhance important channels while suppressing irrelevant ones. Furthermore, based on channel weighting, it focuses on the most important spatial regions in the feature map, significantly enhancing the feature's expressiveness. The process is as follows:

$$V_c = M_c(V) \otimes V, V_s = M_s(V_c) \otimes V_c \tag{10}$$

$$V' = \delta(\mathrm{B}(dropout(V_s))) \tag{11}$$

where V_c represents features selected through channel attention, V_s denotes features selected through spatial attention, and V' is the final feature map output.

E. HYBRID PARALLEL CONVOLUTION BLOCK (HPCB)

The HPCB plays a crucial role in the design of the decoder, featuring four parallel branches for feature extraction. This architecture effectively restores the details and structure of images, with each branch focusing on different types of feature extraction to ensure the capturing of rich local and global information during the decoding process. As shown in **Figure 4**, the first branch consists of a 1×1 convolution followed by Batch Normalization. The 1×1 convolution maintains the spatial dimensions while transforming channel features, enhancing the flexibility of the network. The second branch includes a 3×3 convolution and Batch Normalization [44]; this branch is essential for recovering image details and capturing local features, playing a key role in image reconstruction. The third branch contains global average pooling, a 1×1 convolution, and a Sigmoid function. It extracts global features through global average pooling, which, combined with the recovery of local features, ensures that the generated output possesses both detail and alignment with the overall structure and semantics. Effective feature fusion can generate richer and more comprehensive feature representations, thereby enhancing the quality of the generated results. Therefore, we fuse the output features of the first three branches and apply the non-linear activation function SiLU [46] to the combined output. To further improve the decoding quality while increasing inference speed, the fourth branch implements a Partial Convolution [47]. This allows for selective convolution on a subset of input channels, effectively splitting the input channels into two parts—one half undergoes 3×3 convolution while the other half does not. The results are then concatenated, optimizing computational efficiency and facilitating faster processing of high-resolution images. Finally, the features from the first three branches are fused and then further integrated with the output from the fourth branch, forming the complete HPCB. By incorporating multiple convolution forms, HPCB achieves efficient feature recovery, enhances contextual information, enables effective feature fusion, and improves generation efficiency, significantly boosting performance in image generation and processing tasks.

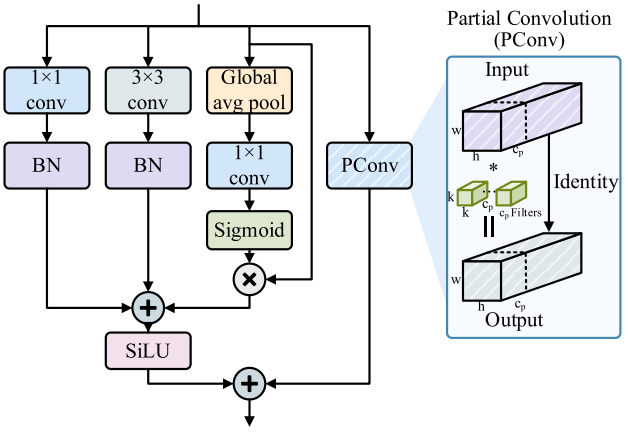


FIGURE 4. Structure of Hybrid Parallel Convolution Block (HPCB).

F. SELF-SUPERVISED LEARNING

Unlike supervised training methods that rely on real depth values, self-supervised learning requires generating self-supervised signals for iterative training, thus transforming the depth estimation task into an image reconstruction task. The depth estimation network generates depth maps D_t based on the input image I_t. Simultaneously, the PoseNet [7], [16], [28] calculates the relative pose $T_{t \to t'}$ from the target image I_t to the adjacent frame $T_{t \to t'}$. The generated depth map D_t and pose information $T_{t \to t'}$ serve as supervisory signals for training.

TABLE 1. The image input resolution was set to 640 × 192 and 1024 × 320. The best and second-best results are highlighted in bold and underlined, respectively. In the table data, "M" represents models trained on monocular videos and pre-trained on ImageNet [51], while "M*" denotes monocular models without pre-training.

Method	Train	Resolution	AbsRel↓	SqRel↓	RMSE↓	RMSElog↓	δ<1.25↑	δ<1.25²↑	δ<1.25³↑
GeoNet[33] (2018)	M	640×192	0.149	1.060	5.567	0.226	0.796	0.935	0.975
DDVO[34] 2018)	M	640×192	0.151	1.257	5.583	0.228	0.810	0.936	0.974
Monodepth2[7] (2019)	M	640×192	0.115	0.903	4.863	0.193	0.877	0.959	0.981
Johnston et al.[27] (2020)	M	640×192	0.111	0.941	4.817	0.189	0.885	0.961	0.981
CADepth-Res18[29] (2021)	M	640×192	0.110	0.812	4.686	0.187	0.882	0.962	0.983
HR-Depth[10] (2021)	M	640×192	0.109	0.792	4.632	0.185	0.884	0.962	0.983
R-MSFM3[35] (2021)	M	640×192	0.114	0.815	4.712	0.193	0.876	0.959	0.981
R-MSFM6[35] (2021)	M	640×192	0.112	0.806	4.704	0.191	0.878	0.960	0.981
MonoFormer[37] (2022)	M	640×192	0.108	0.806	4.594	0.184	0.884	0.963	0.983
Shim and Kim[38] (2023)	M	640×192	0.106	0.739	4.510	0.182	0.890	0.964	**0.984**
Lite-Mono[16] (2023)	M	640×192	0.107	0.765	4.561	0.183	0.886	0.963	0.983
RSANet[8] (2024)	M	640×192	0.106	0.762	4.439	0.185	0.889	0.965	0.983
Sun et al.[39] (2024)	M	640×192	0.104	0.705	4.455	0.179	0.891	0.965	**0.984**
Ours	M	640×192	**0.100**	**0.696**	**4.413**	**0.178**	**0.896**	**0.966**	**0.984**
Monodepth2[7] (2019)	M*	640×192	0.132	1.044	5.142	0.210	0.845	0.948	0.977
R-MSFM3[35] (2021)	M*	640×192	0.128	0.965	5.019	0.207	0.853	0.951	0.977
R-MSFM6[35] (2021)	M*	640×192	0.126	0.944	4.981	0.204	0.857	0.952	0.978
Lite-Mono[16] (2023)	M*	640×192	0.121	0.876	4.918	0.199	0.859	0.953	0.980
Ours	M*	640×192	**0.114**	**0.832**	**4.702**	**0.193**	**0.873**	**0.958**	**0.981**
Monodepth2[7] (2019)	M	1024×320	0.115	0.882	4.701	0.190	0.879	0.961	0.982
R-MSFM3[35] (2021)	M	1024×320	0.112	0.773	4.581	0.189	0.879	0.960	0.982
R-MSFM6[35] (2021)	M	1024×320	0.108	0.748	4.470	0.185	0.889	0.963	0.982
HR-Depth[10] (2021)	M	1024×320	0.106	0.755	4.472	0.181	0.892	0.966	0.984
Lite-Mono[16] (2023)	M	1024×320	0.102	0.746	4.444	0.179	0.896	0.965	0.983
Ours	M	1024×320	**0.097**	**0.681**	**4.332**	**0.176**	**0.902**	**0.967**	**0.984**

1) PHOTOMETRIC CONSISTENCY LOSS

Based on [31], we set the learning objective to minimize the reconstruction loss L_p between the original image I_t and the synthesized target image $I_{t'}$, L_p is defined:

$$L_p = \min_{t'} F\left(I_t, I_{t'\to t}\right) \tag{12}$$

where F in (12) denotes the photometric reconstruction error, and $I_{t'\to t}$ represents the warp result from image $I_{t'}$ to image I_t.

$$I_{t'\to t} = I_{t'}\langle proj(D_t, T_{t\to t'}, K)\rangle \tag{13}$$

where proj () refers to the two-dimensional coordinates generated by projecting the depth map D_t onto the image I_t, and $\langle\,\rangle$ represents sampling operator K is the camera's intrinsics.

$$F\left(I_a, I_b\right) = \frac{\alpha}{2}\left(\left(1 - SSIM\left(I_a, I_b\right)\right) + \left(1-\alpha\right)\left\|I_a - I_b\right\|\right) \tag{14}$$

where I_a represents the original image, and I_b denotes the reconstructed image obtained via (13). $F(I_a, I_b)$ signifies the weighted sum between the Structural Similarity Index Measure (SSIM) [48] and the intensity difference term. The parameter α is set to 0.85 [7].

2) MINIMUM PHOTOMETRIC LOSS

To address occlusions in the source image, the minimum photometric loss for each pixel is calculated between the forward and backward offsets between adjacent frames [7].

$$L_{SS}\left(I_s, I_t\right) = \min_{i\in[-1,1]} F\left(I_a, I_b\right) \tag{15}$$

where "1" and "−1" indicate the forward and backward adjacent frames, respectively.

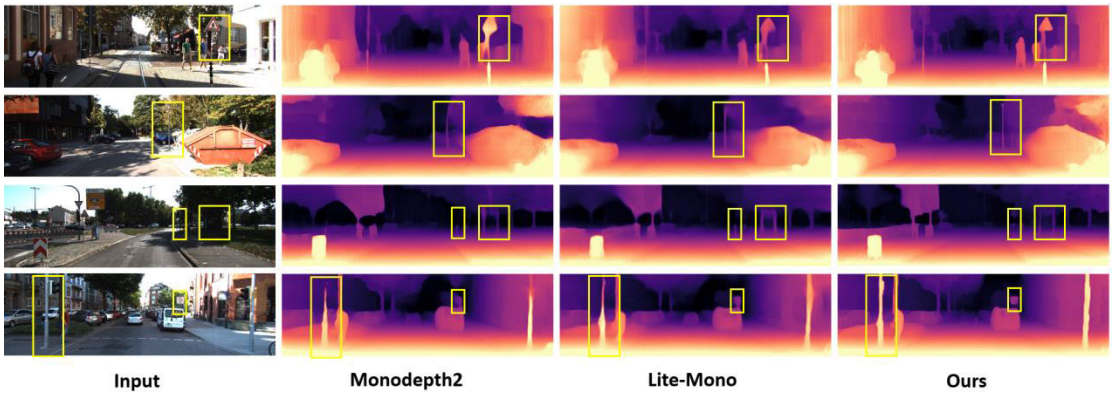


FIGURE 5. Qualitative results on KITTI dataset.

3) AUTOMATIC MASK

To ensure that objects are static relative to the camera, we use an automatic mask [7] to filter out dynamic objects in the image, denoted by u.

$$u = \min_{i\in[-1,1]} L_{SS}\left(I_s, I_t\right) > \min_{i\in[-1,1]} L_{SS}\left(I_a, I_b\right) \tag{16}$$

4) SMOOTHNESS LOSS

Based on [7] and [11], we utilize an edge-aware smoothness loss to prevent arbitrary shrinkage of the inverse depth maps.

$$L_{smooth} = \left|\partial_x d_t^*\right| \cdot e^{-\left|\partial_x I\right|} + \left|\partial_y d_t^*\right| \cdot e^{-\left|\partial_y I\right|} \tag{17}$$

where $d_t^* = \frac{d_t}{\hat{d}_t}$ represents the average normalized inverse depth.

5) TOTAL LOSS

The total loss L is composed of the image reconstruction loss $uL_{SS}\left(I_s, I_t\right)$ and the smoothness loss $\lambda \cdot L_{smooth}$, expressed as:

$$L = \frac{1}{3}\sum_{i=1}^{3}\left(uL_{SS}\left(I_s, I_t\right) + \lambda \cdot L_{smooth}\right) \tag{18}$$

where λ denotes the weight of the smoothness term, set to $1e^{-3}$.

IV. EXPERIMENTS

A. DATASETS

KITTI dataset [18] is a crucial benchmark in the fields of computer vision and autonomous driving, comprising stereo vision data from 61 scenes, and is widely used for tasks such as depth estimation, stereo vision, object detection, and tracking. To train and evaluate the proposed method, we utilize the image segmentation scheme defined by Eigen et al. [49], which includes 39,180 monocular training images, 4,424 validation images, and 697 test images. Additionally, we set the principal point at the center of the images and averaged the focal lengths of all images in the KITTI dataset to ensure that the same intrinsic camera parameters [7] are used during training. According to the official guidelines of the KITTI dataset, the maximum depth is set to 80 meters.

Make3D dataset [19] consists of 134 outdoor scene test images and is commonly used to assess the generalization ability of monocular depth estimation frameworks. We tested our model, trained on the KITTI dataset, on Make3D images and provided corresponding depth map visualizations.

NYU Depth V2 dataset [20] is a collection of indoor video scenes captured using Microsoft Kinect's RGB and depth perception cameras. It encompasses 1449 pairs of detailed labeled RGB and depth images, representing the diversity of the dataset, including various indoor scenes, lighting conditions, and object configurations. This selection ensures that our evaluation results comprehensively reflect the model's performance across different scenarios.

Cityscapes dataset [21] encompasses 5000 high-resolution images of urban driving scenarios, featuring a wealth of moving objects and intricate scene variations. To gauge the generalizability of our model, we utilized a model pretrained on the KITTI dataset to directly generate depth map visualizations for images within the Cityscapes dataset.

B. IMPLEMENTATION DETAILS

In the experiments, our model is implemented based on the PyTorch framework and is trained on a single NVIDIA RTX 4060ti (16GB) with a batch size of 12. The model employs the AdamW optimizer [50], with a weight decay set to 1e-2. Both the pose encoder and depth encoder are pre-trained on ImageNet [51] When using the pre-trained weights, the model is trained for 30 epochs with a cosine learning rate schedule [52], starting from an initial learning rate of 5e-6. For models trained from scratch, the network undergoes training for 35 epochs, with a minimum learning rate set to 5e-4. The training time for the pre-trained model is approximately 22 hours, while the model trained from scratch takes around 35 hours. In our experiments, we adopt the same data augmentation strategies as in previous research [7], [16]. To evaluate the model's effectiveness, we utilize seven widely adopted metrics for depth estimation as proposed by Eigen et al. [22]. These metrics include Abs Rel, Sq Rel, RMSE, RMSE log, $\delta < 1.25$, $\delta < 1.25^2$, and $\delta < 1.25^3$. Lower

TABLE 2. Comparison of generalization on Make3D dataset. Best metrics are in bold.

Method	Abs Rel	Sq Rel	RMSE	RMSE log
DDVO[34] (2018)	0.387	4.720	8.090	0.204
Monodepth2[7] (2019)	0.322	3.589	7.417	0.163
R-MSFM6[35] (2021)	0.334	3.285	7.212	0.169
Lite-Mono[16] (2023)	0.305	3.060	6.981	0.158
EDFNet (ours)	**0.292**	**2.822**	**6.816**	**0.153**

values of Abs Rel, Sq Rel, RMSE, and RMSE log indicate smaller errors and greater accuracy in depth estimation. Conversely, higher values of $\delta < 1.25$, $\delta < 1.25^2$, and $\delta < 1.25^3$ signify a greater proportion of depth estimates that align with the ground truth within the specified error thresholds.

C. KITTI RESULTS

We comprehensively compared our method with other monocular depth estimation methods on the KITTI dataset. We utilized two different resolutions: 640 × 192 and 1024 × 320. In the case of the 640 × 192 resolution, we differentiated between models that employed pre-trained weights and those that did not; the latter approach was aimed at better validating the model's ability to learn features. As shown in **Table 1**, our method demonstrates superior performance across all seven evaluation metrics, outperforming other self-supervised methods at both high-resolution and low-resolution inputs. At the 640 × 192 resolution, EDFNet significantly surpassed the methods proposed by Monodepth2 [7], Lite-Mono [16], and Sun et al. [38], achieving a notable accuracy metric of 0.896 for the challenging $\delta < 1.25$ criterion. Furthermore, at the 1024 × 320 resolution, our model attained an accuracy of 0.902, showcasing a significant advantage that indicates the effectiveness of our model in handling depth information. To illustrate the effectiveness of our proposed solution in depth estimation, we also provided qualitative comparisons across different scenes in the KITTI dataset.

As depicted in **Figure 5**, the depth maps generated by Monodepth2 [7] and Lite-Mono [16] often exhibit blurred edges for objects such as road signs, roadside billboards, and streetlights, making it challenging to distinguish them accurately. In contrast, the depth maps produced by EDFNet clearly delineate the edges of these objects, resulting in more complete contours and a more accurate reflection of their actual positions and shapes. This improvement is attributed to the model's exceptional detail recovery capabilities and global understanding, allowing it to generate depth maps that are both stable and precise.

D. MAKE3D RESULTS

We conducted a detailed experimental validation of our proposed depth estimation method on the Make3D dataset. The purpose was to ensure that our model performs well not only under specific conditions but also maintains its performance and accuracy across various outdoor scenes. As shown in **Table 2** compared to four other classic models, EDFNet achieved the best performance on four key metrics. The Abs Rel metric measures the relative error between predicted depth and ground truth depth, with EDFNet achieving a minimum value of 0.292. This performance surpasses that of other models, such as Monodepth2 [7] (0.322) and Lite-Mono [16] (0.305), demonstrating higher accuracy in depth value predictions. This confirms the effectiveness of our approach in depth estimation tasks.

In **Figure 6**, we provide qualitative results across different scenes in the Make3D dataset. Our comparative analysis shows that our model produces more complete contours for vehicles and streetlights, while accurately reconstructing the shapes of road signs and trees. These results substantiate the superiority of our method, particularly in the detail-capturing capabilities of the depth maps generated by EDFNet. This advantage is largely attributed to the innovative design of our model, which effectively integrates local and global information, allowing it to perceive objects of varying sizes and accurately identify their relative positions to produce more precise depth maps. This capability is especially critical in complex outdoor scenes, as it involves the accurate recognition of objects at different scales and their corresponding depth estimation.

E. NYU DEPTH V2 RESULTS

To further explore the generalization capabilities of our model, we specifically focused on its ability to handle complex indoor scenes during the evaluation process. We transitioned the testing environment from outdoor to indoor settings, covering aspects such as the recognition of small objects, stability under varying lighting conditions, and precise depth estimation of objects at different distances within the scene. Our model utilized weights that were pre-trained on the KITTI dataset and was subsequently evaluated on the NYU Depth V2 dataset. According to the evaluation results presented in **Table 3**, our method outperformed several classic models across all seven key assessment metrics. Notably, in terms of the critical Abs Rel and $\delta < 1.25$ metrics, our model surpassed the recent method Lite-Mono [16] by 0.023 and 0.033, respectively, demonstrating exceptional performance.

Additionally, we provide qualitative comparisons in **Figure 7**, where our method is contrasted with several other approaches. The depth maps clearly illustrate our model's superiority in capturing intricate details within indoor environments. For instance, our method effectively delineates the contours of objects such as doors, computers, and sofas, while also representing distant objects like doors with clarity. In contrast, other models struggle to achieve similar levels of accuracy in these representations. In summary, our experimental results not only validate the model's effectiveness

TABLE 3. Comparison of generalization on NYU Depth V2 dataset. Best metrics are in bold.

Method	Abs Rel	Sq Rel	RMSE	RMSE log	$\delta<1.25$	$\delta<1.25^2$	$\delta<1.25^3$
Monodepth2[7] (2019)	0.355	0.673	1.252	0.373	0.485	0.771	0.907
DynaDepth[55] (2022)	0.335	0.589	1.189	0.361	0.502	0.784	0.916
Lite-Mono[16] (2023)	0.329	0.546	1.169	0.363	0.499	0.781	0.915
EDFNet (ours)	**0.306**	**0.498**	**1.157**	**0.341**	**0.532**	**0.809**	**0.930**

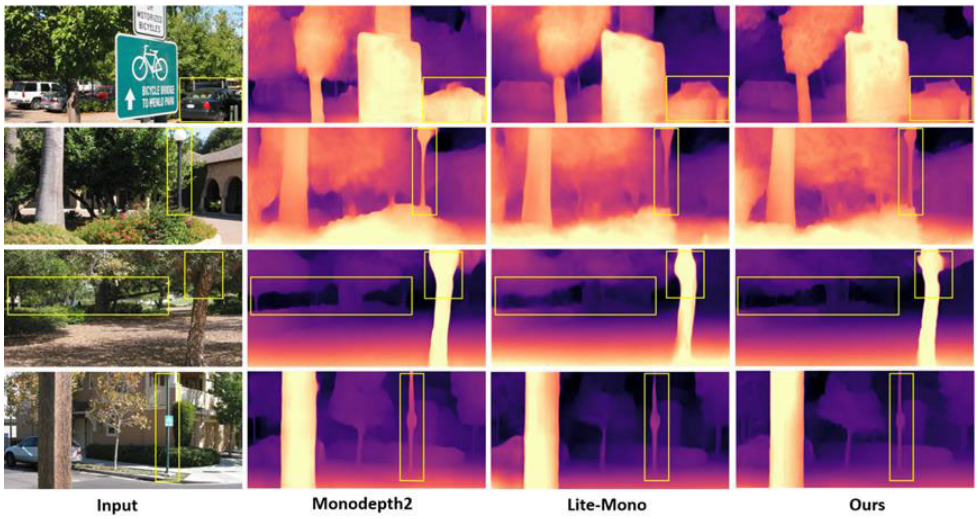


FIGURE 6. Qualitative results on Make3D dataset.

when trained on an outdoor dataset but also highlight its exceptional performance in complex indoor environments.

F. CITYSCAPES RESULTS

To further verify the model's adaptability in more complex and variable scenarios, we conducted generalization experiments on the Cityscapes dataset and provided depth map visualization results, as shown in **Figure 8**. In these results, our model is capable of more clearly recovering the contours and shapes of the cyclist and tree trunks in the image, with the generated depth maps having more defined boundaries. The model effectively captures the edges of objects through refined feature extraction and fusion, generating sharper and more coherent structures in the depth maps. In contrast, Monodepth2 [7] and Lite-Mono [16] methods suffer from blurring and discontinuities in the depth maps, leading to a loss of detail information. Additionally, distant cars are accurately identified and positioned in our depth maps, with the boundaries of parked cars along the roadside being much clearer and the depth information being highly consistent with the actual scene. This is mainly due to the model's excellent performance in handling distant objects. It can effectively use multi-scale feature fusion and contextual information integration to maintain precise perception of faraway objects. On the other hand, Monodepth2 [7] and Lite-Mono [16] methods have deviations in the depth estimation of cars, resulting in significant discrepancies between the depth maps and the actual scene. These results indicate that our model has higher accuracy in handling dynamic objects and complex occlusions in complex scenes.

G. ABLATION EXPERIMENTS ON MODEL STRUCTURES

To thoroughly investigate the effectiveness of each component within the network, we conducted a detailed ablation study aimed at analyzing and quantifying the contribution of each module to the overall performance. The experimental results are presented in **Table 4**. We used the original network as a baseline, systematically removing or modifying certain modules to evaluate their impact on performance. Our focus was particularly on three core components: the Multi-Dimensional Attention Fusion Module (MDAFM), the Adaptive Selective Attention Module (ASAM), and the Hybrid Parallel Convolution Block (HPCB), along with a Skip-connection. These modules collectively represent the innovative aspects of our network architecture, each designed to enhance the model's ability to process depth information and recover details more effectively.

1) THE BENEFITS OF MDAFM

As shown in **Table 4**, the performance of the model without MDAFM (w/o MDAFM) exhibited a decline across all seven key metrics. The Abs Rel, Sq Rel, RMSE, RMSE log,

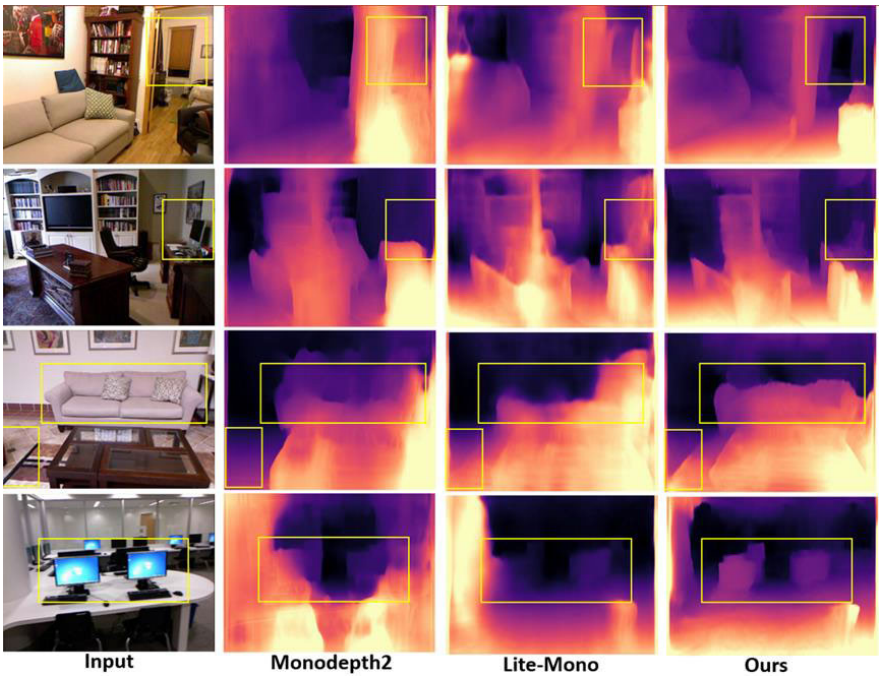


FIGURE 7. Qualitative results on NYU Depth V2 dataset.

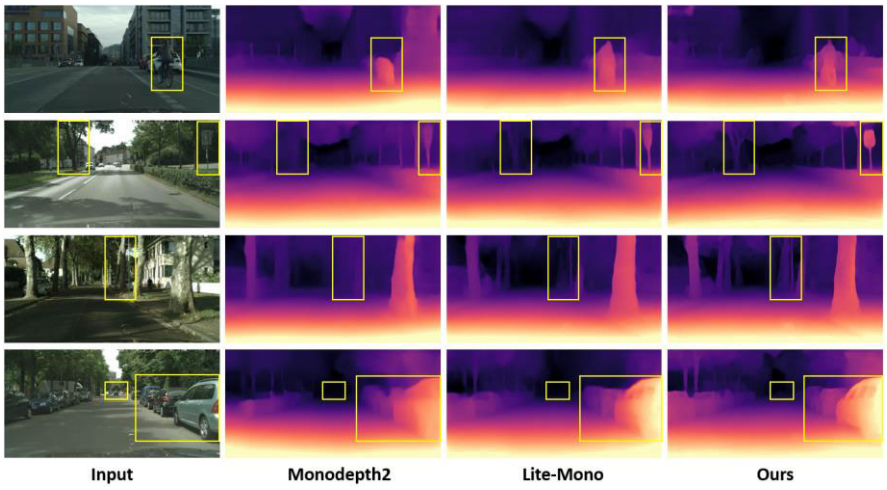


FIGURE 8. Qualitative results on Cityscapes dataset.

$\delta < 1.25$, $\delta < 1.25^2$, and $\delta < 1.25^3$ decreased to 0.104, 0.751, 4.511, 0.180, 0.890, 0.964, and 0.983, respectively. Notably, the $\delta < 1.25$ decreased by 0.006, which is a critical indicator of depth estimation accuracy, highlighting the importance of MDAFM in detail recovery. This is attributed to MDAFM's ability to capture both local and global features at different scales, which facilitates more accurate detail recovery in images. Furthermore, the serial convolution branches of MDAFM further integrate local and global information, providing a more comprehensive understanding of the context.

2) THE BENEFITS OF ASAM

ASAM demonstrates unique advantages in enhancing network performance by providing richer feature representations through the operations of split, fuse, select, and strengthen. The richness of this representation enables the model to capture image details more precisely, particularly in areas involving object edges and textures. As indicated in **Table 4** the model without ASAM (w/o ASAM) showed poorer performance in the Abs Rel and Sq Rel, with values dropping to 0.104 and 0.764, respectively, reflecting reductions of

TABLE 4. Ablation study on model architectures. All the models are trained and tested on KITTI dataset with the input size 640 × 192.

Architecture	Abs Rel	Sq Rel	RMSE	RMSE log	δ<1.25	δ<1.25^2	δ<1.25^3
Baseline	0.107	0.813	4.578	0.183	0.886	0.963	0.983
Baseline + MDAFM	0.105	0.773	4.529	0.181	0.890	0.964	0.983
Baseline + ASAM	0.105	0.760	4.526	0.181	0.889	0.964	0.983
Baseline + HPCB	0.106	0.782	4.542	0.182	0.888	0.963	0.983
Baseline + ASAM + HPCB + Skip-connect (w/o MDAFM)	0.104	0.751	4.511	0.180	0.890	0.964	0.983
Baseline + MDAFM + HPCB + Skip-connect (w/o ASAM)	0.104	0.764	4.499	0.180	0.891	0.964	0.983
Baseline + MDAFM + ASAM + Skip-connect (w/o HPCB)	0.103	0.736	4.473	0.179	0.894	0.965	0.984
Baseline + MDAFM + ASAM + HPCB (w/o Skip-connect)	0.102	0.721	4.445	0.178	0.895	0.966	0.984
Baseline + MDAFM + ASAM + HPCB + Skip-connect (Full Model)	**0.100**	**0.696**	**4.413**	**0.178**	**0.896**	**0.966**	**0.984**

0.004 and 0.068. This confirms the significance of ASAM in reducing the model's error rates.

3) THE BENEFITS OF HPCB

Table 4 reveals that the model without HPCB (w/o HPCB) performed slightly worse than the full model, with the seven key metrics recorded as 0.103, 0.736, 4.473, 0.179, 0.894, 0.965, and 0.984, respectively. This indicates that MDAFM and ASAM have a more pronounced effect on enhancing performance. Notably, the absence of HPCB resulted in only minor reductions of 0.002 and 0.001 in the $\delta < 1.25$ and $\delta < 1.25^2$, while the $\delta < 1.25^3$ remained unchanged. Nevertheless, HPCB plays a crucial role in maintaining the overall performance and efficiency balance of the model. Its efficient feature extraction and fusion capabilities significantly contribute to detail and structural recovery while preserving computational efficiency, which is vital for real-time applications.

4) THE BENEFITS OF SKIP-CONNECTION

The Skip-connection [17] is pivotal within our EDFNet architecture. The performance metrics in **Table 4** underscore the substantial benefits of incorporating Skip-connection, particularly in the Abs Rel, Sq Rel, RMSE, and $\delta < 1.25$. These four metrics improved by 0.002, 0.025, 0.032, and 0.001, respectively, demonstrating the effectiveness of skip connections in maintaining spatial detail integrity. This architectural choice allows for the direct fusion of features from different stages of the network, reinforcing its critical role in achieving state-of-the-art depth estimation within our design.

H. LIMITATION AND DISCUSSION

The current validation is confined to NVIDIA GPUs, which to some extent restricts the performance evaluation of the model on other hardware platforms, such as ARM-based edge devices and AMD GPUs. This limitation may affect the model's wide applicability and compatibility in diverse application scenarios. future work will focus on extending the model to these varied hardware platforms to ensure its outstanding performance in a broader range of environments. Additionally, computational complexity remains a key challenge for the model in real-time applications. The current model architecture, particularly the Multi-Dimensional Attention Fusion Module (MDAFM) and the Adaptive Selective Attention Module (ASAM), demands significant computational resources. This somewhat limits the model's deployment efficiency in resource-constrained environments. Therefore, in our subsequent work, we will focus on optimizing the model architecture to enhance its computational efficiency while ensuring that the model's accuracy is not compromised. In this way, we hope to make the model more suitable for deployment in resource-limited settings and improve its feasibility in practical applications.

Despite the significant success of EDFNet in generating high-quality depth maps, the model still faces some challenges in certain specific scenarios. For instance, under extremely low-light conditions, the model may struggle to accurately recover details due to the lack of sufficient visual cues. Additionally, for objects with highly reflective surfaces (such as mirrors or car windows), the accuracy of depth estimation may be affected due to the difficulty in capturing reliable texture information. These failure cases highlight the necessity for further improvements in the model to address these challenging conditions. Future research will be dedicated to solving these issues to further enhance the model's performance and robustness in complex scenarios.

V. CONCLUSION

This paper presents a novel self-supervised monocular depth estimation network, EDFNet, which significantly enhances the accuracy of depth estimation through the innovative encoder-decoder architecture. The design focuses on three key components: the ASAM, the MDAFM, and the HPCB. These modules work together to improve the network's ability to recover details and understand global context, enabling the generation of high-quality depth maps in both outdoor and

indoor scenes. Extensive experiments on the KITTI dataset have demonstrated the superior performance of EDFNet across different resolutions and pre-training strategies. Compared to existing methods, our approach shows significant advantages in multiple evaluation metrics, particularly in the challenging $\delta < 1.25$ accuracy metric, and excels in detail recovery and edge sharpness. The generalization ability of EDFNet has also been fully verified, with better performance than several advanced methods on the Make3D, NYU Depth V2, and Cityscapes datasets. Additionally, ablation experiments further confirm the importance of each component in improving network performance, proving the effectiveness of our design. In summary, EDFNet not only advances the theoretical development of monocular depth estimation but also holds significant application value in practical scenarios such as robot navigation and autonomous driving. Future work will explore the application of EDFNet to a wider range of tasks and scenarios, and further optimize its computational efficiency for broader practical deployment. We believe that the introduction of EDFNet brings new perspectives to the field of monocular depth estimation and provides a solid foundation for future research.

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SHUN LI is currently pursuing the master's degree with Guangxi University, Nanning, China. His research interest includes monocular depth estimation in computer vision.



CHONGZHENG HUANG was born in 1978. He received the master's degree. He is currently an Associate Professor. His main research interests include machine vision and artificial intelligence.



XIANGZHE LI is currently pursuing the bachelor's degree with Guangxi University, Nanning, China. His research interest includes monocular depth estimation in computer vision.



ZHENGYOU LIANG was born in 1968. He received the Ph.D. degree. He is currently a Professor. His main research interests include computer vision, artificial intelligence, and parallel distributed computing. He is a member of CCF.